

JSON Infrastructure Documentation

This document outlines the high-level structure and details the core components of the "ebd_v7_3_stations-infrastructure-description.json" infrastructure document used to represent a railway network.

High-Level Structure

The infrastructure file is divided into 13 main parts, each addressing a specific aspect of the railway network definition.

Part	Description
<code>nodes</code>	Network junctions, typically switches, and handover gates.
<code>tracks</code>	The physical rail segments connecting nodes.
<code>allocationSections</code>	Sections of track allocated together for train movement.
<code>platforms(leer)</code>	Currently unused, reserved for future extensions.
<code>stoppingLocations</code>	Points where a train can or must stop.
<code>speedConstraints(leer)</code>	Currently unused, reserved for future extensions.
<code>stoppingLocationGroups</code>	Logical groupings of individual stopping locations
<code>segmentProfiles</code>	Directional portion of a track
<code>timingPoints</code>	Spatial & temporal points
<code>virtualNodes</code>	Artificial nodes used for testing and simulation
<code>virtualTracks</code>	Artificial tracks used for testing and simulation
<code>blockSection (leer)</code>	Currently unused, reserved for future extensions.
<code>dpsGroup</code>	Drive Protection Section Groups. Presence in each section is mutually exclusive.

Core Component Details

Nodes

Nodes are general topological points in the railway network. Most nodes represent switches (simplePoint), some represent external entry/exit points (handoverGate), others are simple geometric connection points without any special function.

Each Node contains the following information:

Key	Type	Description
id	UUID	Unique identifier for the node.
coordinate	Object	The geographic X and Y coordinates of the node.
simplePoint/handoverGate	Object	Nodes are either simple Points or handover Gates. Simple Points contain information necessary to treat the Node as a switch. Handover Gates are the sources and sinks, where trains emerge or end.
numericId	Integer	A numerical identifier for the node.

Simple Points

simplePoint objects contain the following:

Key	Type	Description
connections	Array	Defines how TrackEdges converge (map from trackA to trackB). Multiple tracks can be joined. Tracks are represented as edges in the graph.
switchDurationSecond	Integer	Estimated time (in seconds) for the switch to change direction.

Key	Type	Description
allocationGroupId	String	Identifier for the allocation group this switch belongs to.

Example Node

```
{
  "id": "9a9eddbb-d3aa-478f-a728-76369c5a9fe5",
  "coordinate": {
    "x": 1483.0,
    "y": 2069.0
  },
  "simplePoint": {
    "connections": [
      {
        "trackA": "c72312d8-3897-40d8-8108-52ca9d58ceb9",
        "trackB": "ba1817fa-3b89-408e-8852-310d7a0e88f6"
      },
      {
        "trackA": "c72312d8-3897-40d8-8108-52ca9d58ceb9",
        "trackB": "02cb63d9-65eb-48ad-88a8-d4d32fe90b27"
      }
    ]
  },
  "switchDurationSecond": 1,
  "allocationGroupId": "72"
},
"numericId": 77
}
```

Tracks

Tracks represent the physical segments of the railway network.

Each track contains the following information:

Key	Type	Description
id	UUID	Unique identifier for the track.
sourceNodeId	UUID	The ID of the starting node for the track.

Key	Type	Description
targetNodeId	UUID	The ID of the ending node for the track.
shapingPoints	Array	A list of coordinates (x, y) that define the path of the track. The specific method for track bending through these points is not detailed.
lengthMeter	Float	The length of the track in meters.
numericId	Integer	A numerical identifier for the track.

Example Track

```
{
  {
    "id": "a73a6efb-c5f0-4a13-a06d-93129a4b6703",
    "sourceNodeId": "a1f2355e-22b0-4f7c-99f7-c1eef5a919e5",
    "targetNodeId": "1560a194-f631-4df6-b152-9a71c0f81715",
    "shapingPoints": [
      {
        "x": 2306.0,
        "y": 1581.0
      },
      {
        "x": 2426.0,
        "y": 1571.0
      },
      {
        "x": 1986.0,
        "y": 2870.0
      }
    ],
    "lengthMeter": 8789.43,
    "numericId": 10
  },
}
```

Allocation Sections

allocationSections define track areas that must be reserved together so a train can move safely through them. They describe which parts of the track form a single

operational block: if a train enters any part of an allocationSection, the entire section is considered occupied and no other train may enter it. Multiple allocationSections can share the same `allocationGroupID`, forming a larger block that must be allocated as a whole.

Each `allocationSection` contains:

Key	Type	Description
<code>allocationGroupID</code>	String	Identifier for a specific group of track sections allocated together.
<code>extent</code>	Object	object defining the boundaries of the allocation on specific tracks.

Extents

`Extent` objects define the specific boundaries of an allocation on a given track:

Key	Type	Description
<code>trackId</code>	UUID	The ID of the track segment.
<code>referenceNodeId</code>	UUID	The ID of the node from which distances are measured along the track.
<code>distanceClose</code>	Float	The distance (in meters) from the <code>referenceNodeId</code> defining the closer boundary of the allocation.?
<code>distanceFar</code>	Float	The distance (in meters) from the <code>referenceNodeId</code> defining the farther boundary of the allocation.?

Example Allocation Section

```
{
  "extent": {
    "trackId": "eead44ec-6e0c-4193-bcb3-2cfdb5e35b0f",
    "referenceNodeId": "8ac444cc-ac20-4d25-a00b-5a8531f9bb51",
```

```
    "distanceClose": 0.0,  
    "distanceFar": 39.7929959305  
  },  
  "allocationGroupId": 106  
},
```

Stopping Locations

stoppingLocations refer to every point in the network where a train can or is required to stop.

Key	Type	Description
id	UUID	Unique identifier for the stopping location.
targetDirectionNodeId	UUID	The ID of the node defining the forward direction of travel from this location.
position	Object	The specific coordinates or position on the track.
platformID	UUID (likely)	The ID of the platform at this Stopping Location. type "" in each example

Example Stopping Location

```
{  
  "id": "18ee69af-682a-45cb-a5e5-c7e6778ca3af-SL-B",  
  "targetDirectionNodeId": "56f9d1f2-c1ac-4633-ae6b-18b30b1bc1a1",  
  "position": {  
    "trackId": "fa56252c-ef49-478c-be7d-25633756641a",  
    "referenceNodeId": "56f9d1f2-c1ac-4633-ae6b-18b30b1bc1a1",  
    "distanceFromRefNode": 8.499999999999986  
  },  
  "platformId": ""  
},
```

Stopping Locations Groups

logical groupings of individual stopping locations, often corresponding to operational areas like stations or specific track sections.

Each `stoppingLocationsGroup` contains:

Key	Type	Description
<code>id</code>	String	Name of stoppingLocationGroup
<code>stoppingLocations</code>	Object	
<code>borders</code>	Array	The borders of the Group? In the example they are all empty

Stopping Locations

`stoppingLocations` objects contain the following:

Key	Type	Description
<code>trackId</code>	UUID	Id of Track
<code>stoppingLocationId</code>	UUID	ID of the corresponding stopping location

Example Stopping Location Group

```
{
  "id": "Franzensfeld",
  "stoppingLocations": [
    {
      "trackId": "4e36b5fe-95f9-4b9c-951e-7d9421c89640",
      "stoppingLocationId": "7fb0b11d-3ba1-46ae-a2b1-237bcfbf4c26-SL-A"
    },
    {
      "trackId": "0da52289-fd0c-4a9f-b841-bcdbf263c85c",
      "stoppingLocationId": "7fb0b11d-3ba1-46ae-a2b1-237bcfbf4c26-SL-B"
    }
  ]
}
```

```
],  
  "borders": [],  
},
```

Segment Profiles

segmentProfiles describe a specific directional portion of a track. They define the exact start and end points of a segment, always measured towards a given target node.

This allows other components, such as timing points, to refer to consistent and well-defined sections of a track."

Key	Type	Description
trackId	UUID	Unique identifier for the segment.
id	Integer	Integer identifier for the segment.
targetDirectionNodeId	UUID	The node defining the direction in which all distances of this segment are measured.
distanceSegmentStartToTargetNodeInMeters	Float	Distance from the segment's start point to the target node, measured along the track.
distanceSegmentEndToTargetNodeInMeters	Float	Distance from the segment's end point to the target node, measured along the track.

Example Segment Profile

```
{  
  "trackId": "0da52289-fd0c-4a9f-b841-bcdbf263c85c",  
  "id": 281,  
  "targetNodeId": "ff83063e-c03f-4447-8bed-164fa776d01b",  
  "distanceSegmentStartToTargetNodeInMeters": 2883.946,  
  "distanceSegmentEndToTargetNodeInMeters": 0.0  
},
```


Timing Points

`timingPoints` are defined as points on the track where a train can stop (but does not have to).

These points are used to determine when the train should be at a given location.

They contain the following:

Key	Type	Description
<code>trackId</code>	UUID	Unique identifier for the track.
<code>id</code>	Integer	The ID of the timing Point
<code>targetNodeId</code>	UUID	The unique identifier for the target node
<code>distanceToTargetNodeInMeters</code>	Float	Float corresponding to the distance to target node in meters
<code>stoppingLocationId</code>	String	
<code>segmentProfileId</code>	Integer	

```
{
  "trackId": "fa56252c-ef49-478c-be7d-25633756641a",
  "id": 68,
  "targetNodeId": "56f9d1f2-c1ac-4633-ae6b-18b30b1bc1a1",
  "distanceToTargetNodeInMeters": 236.93965517241378,
  "stoppingLocationId": "",
  "segmentProfileId": 4
}
```

Virtual Nodes

`virtualNodes` are nodes that don't exist but are included for test purposes only. Not in our Scope

They contain the following:

Key	Type	Description
<code>id</code>	UUID	Unique identifier for the node.
<code>coordinate</code>	Object	The geographic X and Y coordinates of the node.
<code>numericId</code>	Integer	A numerical identifier for the node.

```
{
  "id":
  "dba2355f-6da6-4702-bf43-819a42cb9ea5_virtual_track_virtual_node",
  "coordinate": {
    "x": 10596.550215349782,
    "y": 7871.154430021751
  },
  "numericId": 117
},
```

Virtual Tracks

`virtualTracks` are tracks that don't exist but are included for test purposes only.
Not in our Scope

They contain the following:

Key	Type	Description
<code>id</code>	UUID	Unique identifier for the track.
<code>sourceNodeId</code>	UUID	The ID of the starting node for the track.
<code>targetNodeId</code>	UUID	The ID of the ending node for the track.
<code>shapingPoints</code>	Array	A list of coordinates (x, y) that define the path of the track. The specific method for track bending through these points is not detailed.

Key	Type	Description
<code>lengthMeter</code>	Float	The length of the track in meters.
<code>numericId</code>	Integer	A numerical identifier for the track.

```
{
  "id": "28c1a9e7-6396-458d-ad33-16a9fc328b05_virtual_track",
  "sourceNodeId":
    "28c1a9e7-6396-458d-ad33-16a9fc328b05_virtual_track_virtual_node",
  "targetNodeId": "28c1a9e7-6396-458d-ad33-16a9fc328b05",
  "shapingPoints": [],
  "lengthMeter": 10000.0,
  "numericId": 153
},
```

DPS Groups

`dpsGroups` are “Drive Protection Section Groups”. They describe a set of track areas where trains could collide if they entered them at the same time.

If a train enters one of these areas, all other areas in the same `dpsGroup` must be blocked so no other train can enter them."

They contain the following:

Key	Type	Description
<code>id</code>	UUID	Unique identifier for the track.
<code>protectedNodeIds</code>	array<UUID>	The IDs of the starting node for the track.
<code>sections</code>	array<Object>	One possible movement area inside the <code>dpsGroup</code> that must be treated as occupied as soon as a train enters it.

Key	Type	Description
dependencyType	String	Defines how the individual trackSections within the same dpsGroup relate to each other.

Example dpsGroup:

```
{
  "id": "8bcfd5b1-33f6-3e6b-bd22-2eadc6e78283",
  "protectedNodeIds": [
    "5e72291c-9820-40df-b903-4c8dddb925eb",
    "45832650-1cff-448e-9d27-484d2b49efdb",
    "5efb2c10-fa94-4324-bae4-95f017d397f4",
    "a9645498-a0e6-4651-91b3-e87b12541faa"
  ],
  "sections": [
    {
      "id": "5dc13f36-b7f9-33d9-aa66-69747ccc8450",
      "trackSections": [
        {
          "trackId": "bfdfc27c-b62a-420d-a02a-7c03923b379b",
          "referenceNodeId": "5e72291c-9820-40df-b903-4c8dddb925eb",
          "coordinateStart": 27.988999999999997,
          "coordinateEnd": 29.999999999999993
        }
      ]
    },
    {
      "id": "f1ea1ce6-8f26-3a1c-9489-0c7d756c099a",
      "trackSections": [
        {
          "trackId": "66fec6af-d73f-4087-bb55-bb0ad317e89b",
          "referenceNodeId": "a9645498-a0e6-4651-91b3-e87b12541faa",
          "coordinateStart": 26.830999999999996,
          "coordinateEnd": 30.000000000000007
        }
      ]
    }
  ],
  "dependencyType":
  "DRIVE_PROTECTION_SECTION_GROUP_DEPENDENCY_EXCLUSIVE"
},
```